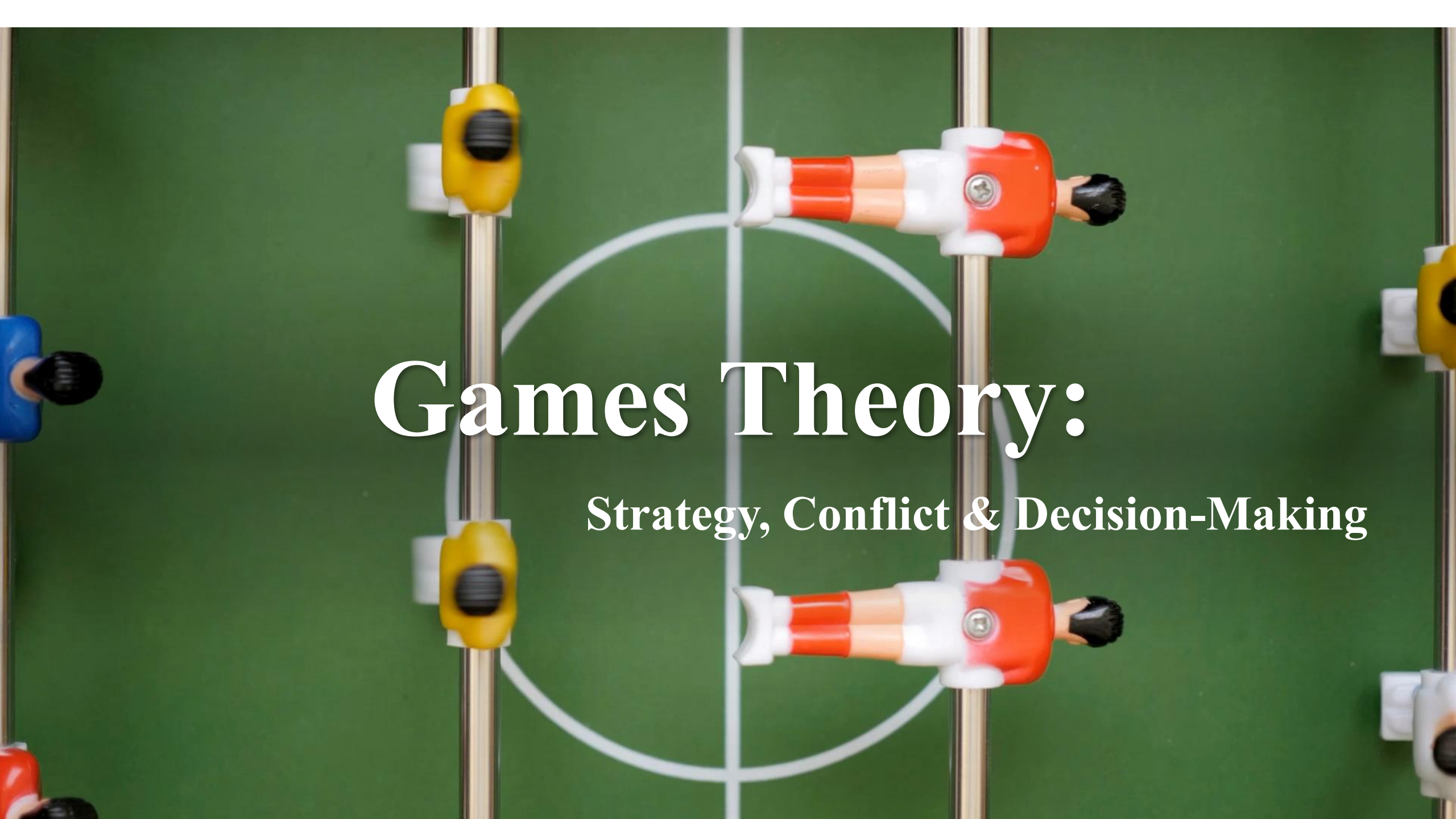


OPERATIONS RESEARCH



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Games Theory:

Strategy, Conflict & Decision-Making

Example: A Dice Game

- Two players A and B each roll a die.
- The sum of the two rolls determines the payoff:
 - If **Sum is Odd** → **A Wins** : Player B pays Player A the value of the sum.
 - If **Sum is Even** → **B Wins**: Player A pays Player B the value of the sum (recorded as negative for A).

Example:

- A throws 1, B throws 6 → sum = 7 (odd) → A receives Rs 7.
- A throws 2, B throws 4 → sum = 6 (even) → A pays Rs 6, recorded as -6.

The Payoff Matrix:

A\B	1	2	3	4	5	6
1	-2	3	-4	5	-6	7
2	3	-4	5	-6	7	-8
3	-4	5	-6	7	-8	9
4	5	-6	7	-8	9	-10
5	-6	7	-8	9	-10	11
6	7	-8	9	-10	11	-12

- This 6×6 matrix is a **rectangular game**.
- Each row is a strategy for A (values 1–6)
- Each column is a strategy for B.
- The entry a_{ij} is the amount **B pays to A**.
- A negative entry means A pays B.

Key Definitions



Game:

A set of rules followed by players with **conflicting objectives**.

Game theory (introduced by von Neumann): Studies rational decision-making under conflict.



2-Person Zero-Sum Game:

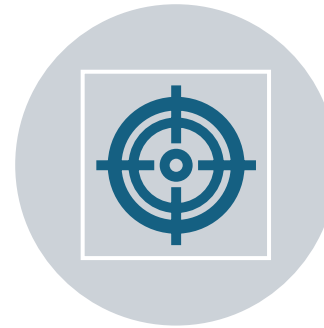
A game where one player's gain exactly equals the other's loss.

Total received by both players at each play = $a_{ij} + (-a_{ij}) = 0$.



Rectangular Game:

A two-player game where A has m strategies and B has n strategies, represented by an $m \times n$ **payoff matrix** $[a_{ij}]$.



Strategy:

A predetermined rule dictating which course of action a player takes at each turn, potentially based on prior moves and the opponent's behavior.



Pure Strategy vs. Mixed Strategy

Pure Strategy:

- Each player plays **exactly one strategy** per game.
- Deterministic.
- Works when a saddle point exists.

Mixed Strategy:

- Each player assigns **probabilities** to their strategies.
- A player A with m strategies, is associated with probabilities $(x_1, x_2, \dots, x_m)$ where each $x_i \geq 0$ and $\sum x_i = 1$.
- Used when no saddle point exists.
- Always yields a saddle value.

Saddle Point

- In a pure strategy game, each player reasons defensively before choosing their strategy.
- If $\text{Max}_i(\text{Min}_j a_{ij}) = \text{Min}_j(\text{Max}_i a_{ij}) = a_{pq}$, then cell (p, q) is a **saddle point** and a_{pq} is the **value of the game**.
- There may be multiple saddle points, but the saddle value is always unique.

Example:

Consider the following 3×3 payoff matrix. Does a saddle point exist?

A \ B	B ₁	B ₂	B ₃
A ₁	2	3	4
A ₂	7	3	-1
A ₃	-2	3	8

- Row Minimums (Player A):
Row 1: $\min(2, 3, 4) = 2$
Row 2: $\min(7, 3, -1) = -1$
Row 3: $\min(-2, 3, 8) = -2$
 $\text{Max}_i(\text{Min}_j a_{ij}) = \text{Max}(2, -1, -2) = 2$
- Column Maximums (Player B):
Col 1: $\max(2, 7, -2) = 7$
Col 2: $\max(3, 3, 3) = 3$
Col 3: $\max(4, -1, 8) = 8$
 $\text{Min}_j(\text{Max}_i a_{ij}) = \text{Min}(7, 3, 8) = 3$
- Since $2 \neq 3$, **no saddle point exists**.
- This game must be solved using mixed strategies.

Saddle Value for Mixed Strategies

- **Expectation Function:** For an $m \times n$ game with payoff matrix $[a_{ij}]$, probability vectors $X = (x_1, \dots, x_m)$ and $Y = (y_1, \dots, y_n)$, the expected payoff is:

$$E(X, Y) = \sum_{i=1}^m \sum_{j=1}^n a_{ij} x_i y_j$$

- **Saddle Condition:** A point (X_0, Y_0) is a saddle point if:

$$E(X, Y_0) \leq E(X_0, Y_0) \leq E(X_0, Y) \text{ for all } X \in S_m, Y \in S_n.$$

NOTE: Every finite $m \times n$ game played with mixed strategies **always has a saddle value** — even when no pure strategy saddle point exists.

Solution of 2×2 Game without Saddle Point (Mixed Strategy Method)

	B_1	B_2
A_1	a_{11}	a_{12}
A_2	a_{21}	a_{22}

such that
no saddle point
exists.

For a 2×2 game with no saddle point, assign probabilities (x_1, x_2) to A's strategies and (y_1, y_2) to B's strategies.

Let v be the value of the game.

Player A : Maximize v

A wants his expected payoff to be **at least v** regardless of B's move:

- If B plays B_1 : $a_{11}x_1 + a_{21}x_2 \geq v$
- If B plays B_2 : $a_{12}x_1 + a_{22}x_2 \geq v$
- $x_1 + x_2 = 1, \quad x_1, x_2 \geq 0.$

Player B : Minimize v

B wants his expected payoff to be **at most v** regardless of A's move:

- If A plays A_1 : $a_{11}y_1 + a_{12}y_2 \leq v$
- If A plays A_2 : $a_{21}y_1 + a_{22}y_2 \leq v$
- $y_1 + y_2 = 1, \quad y_1, y_2 \geq 0.$
- Treating constraints as equalities and solving simultaneously yields the optimal probabilities and game value v .

Example: Saddle Point in Game Theory

Find the range of values of p and q which will render (2, 2), a saddle point for the game.

	B_1	B_2	B_3
A_1	4	p	2
A_2	q	5	7
A_3	10	3	9

- Given: (2,2) entry = 5 is a saddle point.
- Thus, 5 should be minimum in row 2 and maximum in column 2.
- $\max(p, 5, 3) = 5$ and $\min(q, 5, 7) = 5$.
- So,

$$p \leq 5 \text{ and } q \geq 5$$

Example: Finding Saddle Point

Find the saddle point, if any, and also give the saddle value for the game.

	$j = 1$	$j = 2$	$j = 3$	$j = 4$
$i = 1$	2	2	2	2
$i = 2$	1	2	3	1

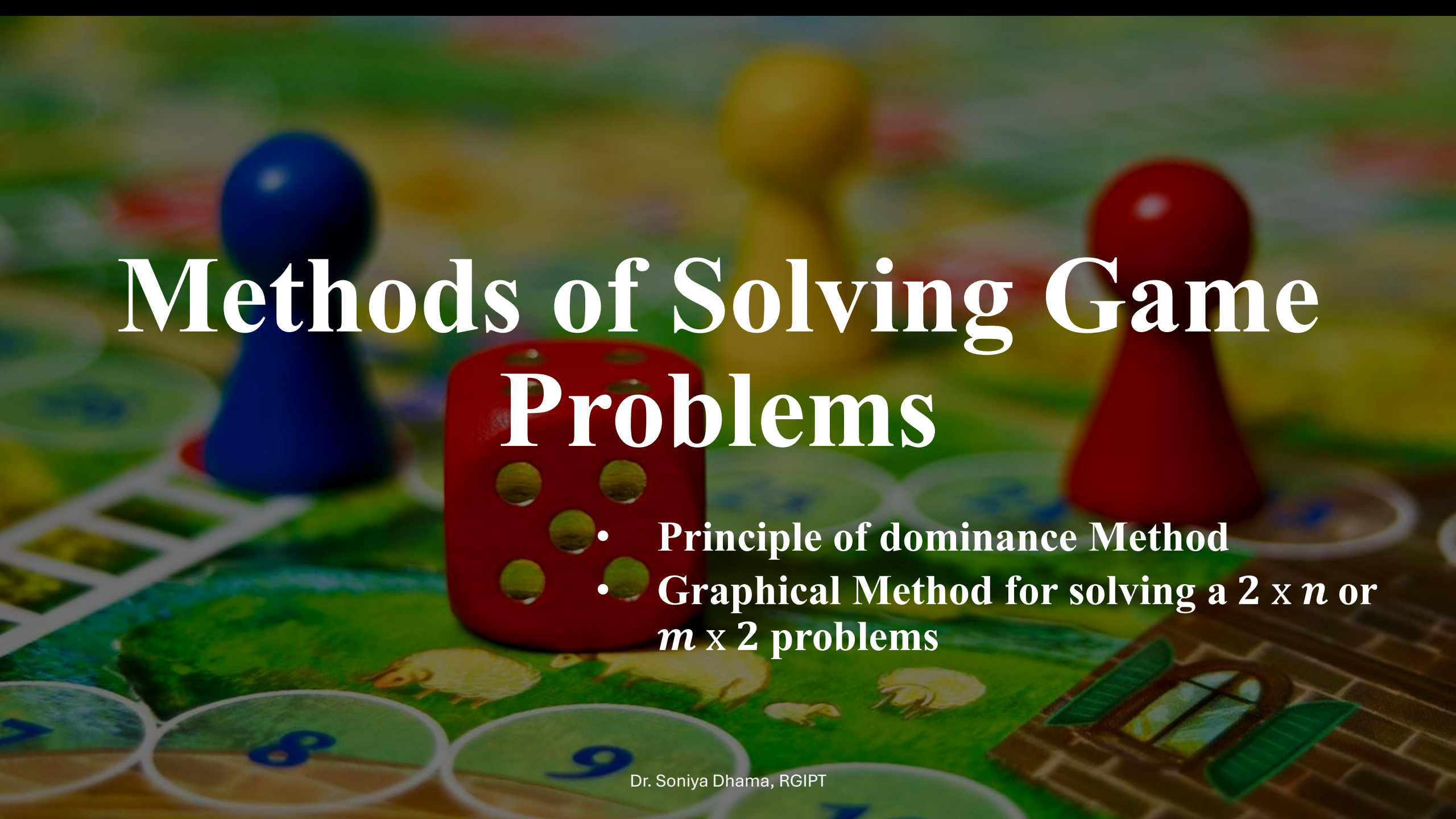
- First, mark the Min (a_{ij}) for $i = 1, 2$ with circles:

	$j = 1$	$j = 2$	$j = 3$	$j = 4$
$i = 1$	2	2	2	2
$i = 2$	1	2	3	1

- Then, mark Max (a_{ij}) for $j = 1, 2, 3, 4$, with squares

	$j = 1$	$j = 2$	$j = 3$	$j = 4$
$i = 1$	2	2	2	2
$i = 2$	1	2	3	1

- $\text{Max}_i(\text{Min}_j a_{ij}) = 2$
- $\text{Min}_j(\text{Max}_i a_{ij}) = 2$
- Saddle value is 2.
- There are three saddle points: (1, 1), (1, 2), (1, 4).

A board game with a green field, a red die, and blue and red pawns.

Methods of Solving Game Problems

- Principle of dominance Method
- Graphical Method for solving a $2 \times n$ or $m \times 2$ problems

Principle of Dominance (6 Rules)

In a 2-person zero-sum game, Player A **maximizes** while Player B **minimizes** the payoff. The payoff matrix can often be simplified before solving.

Dominance rules let us **eliminate rows and columns** that a rational player would never choose — shrinking the problem without losing the optimal solution.

- Dominated strategies are assigned **zero probability** in the optimal mixed strategy.

Rule I: Row Dominance — Direct

The Condition

Row p dominates row q if:

$$a_{pj} \geq a_{qj}, \quad \text{for all } j = 1, \dots, n$$

Player A, as the **maximizer**, will never play a dominated row. The optimal strategy sets $x_q = 0$ and row q is deleted.

IDENTIFY

Find a row where every entry is $\geq$ every corresponding entry of another row.

ELIMINATE

Set the dominated row's probability to zero and remove it from the matrix.

SIMPLIFY

Continue solving the reduced matrix.

Rule II: Column Dominance — Direct

The Condition

Column r dominates column s if:

$$a_{ir} \geq a_{is}, \text{ for all } i = 1, \dots, m$$

Player B is the **minimizer** (a higher-valued column is *worse*), so they avoid the dominating column. The optimal strategy sets $y_r = 0$ and **column r is deleted**.

IDENTIFY

Find a column where every entry is $\geq$ every corresponding entry of another row.

ELIMINATE

Set the dominating column's probability to zero and remove it from the matrix.

SIMPLIFY

Continue solving the reduced matrix.

Rule I vs. Rule II:

A deletes **dominated** rows (lower values). B deletes **dominating** columns (higher values).

Rule III:

Convex Row Combination Dominates a Row

If a convex combination of any set of rows dominates another row q :

$$\sum_k \lambda_k a_{kj} \geq a_{qj}, \quad \text{for all } j, \quad \lambda_k \geq 0, \quad \sum_k \lambda_k = 1$$

Then Player A (maximizer) assigns zero probability to row q and deletes it from the matrix.

Mix rows

Take a weighted average of existing rows.

Dominate & delete

If the mix beats another row, eliminate that row.

Rule IV:

Convex Column Combination Dominates a Column

If a single column s is dominated by a convex combination of other columns:

$$\sum_k \lambda_k a_{ik} \leq a_{is}, \quad \text{for all } i, \quad \lambda_k \geq 0, \quad \sum_k \lambda_k = 1$$

Then Player B (minimizer) would prefer the combination over column s — so **column s is deleted** and assigned zero probability.

B minimizes. A lower-valued combination is better for B, making the standalone dominating column undesirable.

Practical Tip

Try averaging two columns and compare entry-by-entry to the candidate column for deletion.

Ambiguous Combinations — Try Each & Pick Best

When it's unclear which member of the combination to remove, these rules provide a decision procedure:

RULE V — ROWS

A convex combination of rows dominates another row.

Delete **each row of the combination one at a time**, solve the resulting game, and select whichever deletion yields the maximum value of v (saddle value).

RULE VI — COLUMNS

A convex combination of columns is dominated by another column.

Delete **each column of the combination one at a time**, solve the resulting game, and select whichever deletion yields the minimum value of v (saddle value).

Rules V and VI are trial-based: solve multiple sub-problems and pick the optimal result — max for A, min for B.

Example 1

Use the relation of dominance to solve the game, whose pay-off matrix is

	B ₁	B ₂	B ₃	B ₄
A ₁	3	2	4	0
A ₂	3	4	2	4
A ₃	4	2	4	0
A ₄	1	4	0	8

where A_1, A_2, A_3, A_4 are strategies of player A and B_1, B_2, B_3, B_4 are strategies of player B.

- Let (x_1, x_2, x_3, x_4) be the optimal strategy of A and (y_1, y_2, y_3, y_4) be the optimal strategy of B.
- A_3 dominates A_1 , and A is a row (maximizing) player.
- Associate zero probability with the dominated row ($x_1 = 0$).
- Delete the first row.

	B ₁	B ₂	B ₃	B ₄
A ₂	3	4	2	4
A ₃	4	2	4	0
A ₄	1	4	0	8

- B_1 dominates B_3 , and B is a column (minimizing) player.
- Associate zero probability with the dominating column ($y_1 = 0$).
- Delete the first column.

	B ₂	B ₃	B ₄
A ₂	4	2	4
A ₃	2	4	0
A ₄	4	0	8

- $B_2 > \frac{1}{2}B_3 + \frac{1}{2}B_4$
- Player B being a minimizing player.
- Will associate zero probability with B_2 ($y_2 = 0$).
- B_2 is deleted.

	B ₃	B ₄
A ₂	2	4
A ₃	4	0
A ₄	0	8

- $A_2 = \frac{1}{2}A_3 + \frac{1}{2}A_4$
- Player A being a maximizing player.
- Will associate zero probability with A_2 ($x_2 = 0$).
- A_2 is deleted.

		y ₃	y ₄
		B ₃	B ₄
x ₃	A ₃	4	0
x ₄	A ₄	0	8

- Let v be the value of the game.
- A will always like his expectation to exceed v corresponding to any strategy that B plays.

- A's game is:

Max v
subject to

$$x_3 + x_4 = 1, \quad x_3, x_4 \geq 0$$

$$0x_3 + 8x_4 \geq v$$

$$4x_3 + 0x_4 \geq v$$

- B's game is :

Min v
subject to

$$y_3 + y_4 = 1, \quad y_3, y_4 \geq 0$$

$$0 \cdot y_3 + 8y_4 \leq v$$

$$4y_3 + 0y_4 \leq v$$

- Taking the constraints as equality, we have for player A:

$$x_3 = v = 8x_4 \text{ or } x_3 = 2x_4$$

- Also $x_3 + x_4 = 1$, implies

$$3x_4 = 1$$

- Thus, $x_3 = \frac{2}{3}$ and $x_4 = \frac{1}{3}$.

- Similarly, we have for player B:

$$4y_3 = v = 8y_4 \text{ or } y_3 = 2y_4$$

- Also, $y_3 + y_4 = 1$, implies

$$3y_4 = 1$$

- Thus, $y_3 = \frac{2}{3}$ and $y_4 = \frac{1}{3}$.

- Therefore, the optimal strategy for player A and B are:

$$(y_1, y_2, y_3, y_4) = \left(0, 0, \frac{2}{3}, \frac{1}{3}\right)$$

$$(x_1, x_2, x_3, x_4) = \left(0, 0, \frac{2}{3}, \frac{1}{3}\right)$$

and

Example 2

Use the relation of Dominance to solve the game whose pay-off matrix is given by:

	B ₁	B ₂	B ₃	B ₄	B ₅	B ₆
A ₁	4	3	1	3	2	2
A ₂	4	3	7	-5	1	2
A ₃	4	3	4	-1	2	2
A ₄	4	3	3	-2	2	2

- Let (x_1, x_2, x_3, x_4) be the optimal strategy of A and $(y_1, y_2, y_3, y_4, y_5, y_6)$ be the optimal strategy of B.
- B₁ dominates B₂ $\Rightarrow y_1 = 0$.
- Delete B₁.

	B ₂	B ₃	B ₄	B ₅	B ₆
A ₁	3	1	3	2	2
A ₂	3	7	-5	1	2
A ₃	3	4	-1	2	2
A ₄	3	3	-2	2	2

- B₂ dominates B₆ $\Rightarrow y_2 = 0$. Delete B₂.

	B ₃	B ₄	B ₅	B ₆
A ₁	1	3	2	2
A ₂	7	-5	1	2
A ₃	4	-1	2	2
A ₄	3	-2	2	2

- B₆ dominates B₅ $\Rightarrow y_6 = 0$. Delete B₆.

	B ₃	B ₄	B ₅
A ₁	1	3	2
A ₂	7	-5	1
A ₃	4	-1	2
A ₄	3	-2	2

- A₃ dominates A₄ $\Rightarrow x_4 = 0$.
- Delete A₄.

	B ₃	B ₄	B ₅
A ₁	1	3	2
A ₂	7	-5	1
A ₃	4	-1	2

- $B_5 > \frac{1}{2}B_3 + \frac{1}{2}B_4$
- $y_5 = 0$. Delete B₅.

	B ₃	B ₄
A ₁	1	3
A ₂	7	-5
A ₃	4	-1

- $A_3 = \frac{1}{2}A_1 + \frac{1}{2}A_2$
- Delete A₃

		y ₃	y ₄
		B ₃	B ₄
x ₁	A ₁	1	3
x ₂	A ₂	7	-5

- **A's game is:**

$$\begin{aligned} & \textbf{Max } v \\ & \text{subject to} \\ & \quad x_1 + 7x_2 \geq v \\ & \quad 3x_1 - 5x_2 \geq v \\ & \quad x_1 + x_2 = 1, \quad x_1, x_2 \geq 0 \end{aligned}$$

- **B's game is :**

$$\begin{aligned} & \textbf{Min } v \\ & \text{subject to} \\ & \quad y_3 + 3y_4 \leq v \\ & \quad 7y_3 - 5y_4 \leq v \\ & \quad y_3 + y_4 = 1, \quad y_3, y_4 \geq 0 \end{aligned}$$

- **Solving for A:**

$$\begin{aligned} & x_1 + 7x_2 = v = 3x_1 - 5x_2 \\ & \text{or} \quad x_1 = 6x_2 \\ & x_1 + x_2 = 1 \text{ implies} \\ & \quad x_1 = \frac{6}{7}, \quad x_2 = \frac{1}{7} \end{aligned}$$

- **Solving for B:**

$$\begin{aligned} & y_3 + 3y_4 = v = 7y_3 - 5y_4 \\ & \text{or} \quad y_3 = \frac{6}{4}y_4 \\ & y_3 + y_4 = 1 \text{ implies} \\ & \quad y_3 = \frac{4}{7}, \quad y_4 = \frac{3}{7} \end{aligned}$$

The optimal strategy for player A is :

$$(x_1, x_2, x_3, x_4) = \left(\frac{6}{7}, \frac{1}{7}, 0, 0\right)$$

and the saddle value of the game is:

$$v = \frac{13}{7}.$$

The optimal strategy for player B is :

$$(y_1, y_2, y_3, y_4, y_5, y_6) = \left(0, 0, \frac{4}{7}, \frac{3}{7}, 0, 0\right)$$

and the saddle value of the game is:

$$v = \frac{13}{7}.$$

Example 3

Solve by principle of dominance, the following game:

	B_1	B_2	B_3
A_1	2	-2	3
A_2	-3	5	-1
A_3	1	3	2

- Let (x_1, x_2, x_3) and (y_1, y_2, y_3) be the optimal strategies of A and B respectively.
- No row dominates any other row, and no column dominates any other column.
- $A_3 > \frac{1}{2}A_1 + \frac{1}{2}A_2$
- Apply Rule V of dominance: Each row of the combination is deleted one at a time and solve the game, whichever gives the maximum value of v is the optimal solution.

Option 1:

- Associate $x_1 = 0$ and Delete A_1

	B ₁	B ₂	B ₃
A ₂	-3	5	-1
A ₃	1	3	2

- $B_2 > B_1$
- Associate $y_2 = 0$ and delete B_2

	B ₁	B ₃
A ₂	-3	-1
A ₃	1	2

- $B_3 > B_1$
- Associate $y_3 = 0$ and delete B_3

B₁

A ₂	-3
A ₃	1

- $A_3 > A_2$
- Associate $x_2 = 0$ and Delete A_2

B₁

A ₃	1
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- The strategy of player A is (0,0,1) and for player B is (1,0,0).
- Value of the game is $v = 1$.

Option 2:

- Associate $x_2 = 0$ and Delete A_2

	B ₁	B ₂	B ₃
A ₁	2	-2	3
A ₃	1	3	2

- $B_3 > B_1$
- Associate $y_3 = 0$ and delete B_3

	B ₁	B ₂
A ₁	2	-2
A ₃	1	3

- Solving for A:

$$2x_1 + x_3 = v = -2x_1 + 3x_2$$

$$\text{or } 2x_1 = x_2$$

$$x_1 + x_3 = 1 \text{ implies}$$

$$x_1 = \frac{1}{3}, \quad x_2 = \frac{2}{3}$$

- Solving for B:

$$2y_1 - 2y_2 = v = y_1 + 3y_2$$

$$\text{or } y_1 = 5y_2$$

$$y_1 + y_2 = 1 \text{ implies}$$

$$y_1 = \frac{5}{6}, \quad y_2 = \frac{1}{6}$$

- The strategy of player A is $(\frac{1}{3}, 0, \frac{2}{3})$ and for player B is $(\frac{5}{6}, \frac{1}{6}, 0)$.
- Value of the game is $v = \frac{4}{3}$.
- The strategy in Option 2 is optimal.**

Equivalence between Game Problem and LPP

- Every game problem can be reduced to a primal-dual pair.
- Solving the game problem is to solve an LPP.
- Solving a an LPP is to solve a games' problem.

Example

Reduce the given game to an LPP and then solve by simplex method.

	B_1	B_2	B_3	B_4
A_1	2	3	0	-2
A_2	1	0	3	2

Step 1: Add a suitable number to all the entries, so that the value of the game (v) becomes strictly positive.

- Here that suitable number is 3.
- Add 3 to all the entries we get

	B_1	B_2	B_3	B_4
A_1	5	6	3	1
A_2	4	3	6	5

Step 2: Associating probabilities to the strategies of A:

- Let (x_1, x_2) be the strategy for A.
- The game for player A is to find x_1, x_2, v

subject to

$$5x_1 + 4x_2 \geq v$$

$$6x_1 + 3x_2 \geq v$$

$$3x_1 + 6x_2 \geq v$$

$$x_1 + 5x_2 > v$$

$$x_1 + x_2 = 1$$

$$x_1, x_2 \geq 0$$

Step 3: Associating probabilities to the strategies of B:

- Let (y_1, y_2, y_3, y_4) be the strategy for B.
- The game for player B is to find y_1, y_2, y_3, y_4, v

subject to

$$5y_1 + 6y_2 + 3y_3 + y_4 \leq v$$

$$4y_1 + 3y_2 + 6y_3 + 5y_4 \leq v$$

$$y_1 + y_2 + y_3 + y_4 = 1$$

$$y_1, y_2, y_3, y_4 \geq 0$$

Step 4: Dividing each of the constraints obtained in **step 2** and **step 3** by v and writing

$$\frac{x_i}{v} = x'_i, \quad \frac{y_j}{v} = y'_j \text{ for } i = 1, 2, j = 1, 2, 3, 4$$

the constraints can be re-written as

$$5x'_1 + 4x'_2 \geq 1$$

$$6x'_1 + 3x'_2 \geq 1$$

$$3x'_1 + 6x'_2 \geq 1$$

$$x'_1 + 5x'_2 > 1$$

$$x'_1 + x'_2 = \frac{1}{v}, \quad x'_1, x'_2 \geq 0$$

$$5y'_1 + 6y'_2 + 3y'_3 + y'_4 \leq 1$$

$$4y'_1 + 3y'_2 + 6y'_3 + 5y'_4 \leq 1$$

$$y'_1 + y'_2 + y'_3 + y'_4 = \frac{1}{v}$$

$$y'_1, y'_2, y'_3, y'_4 \geq 0$$

- A is a maximizing player
- He wishes to maximize v , i.e.,

$$\text{Min } \frac{1}{v} = x'_1 + x'_2$$

subject to

$$5x'_1 + 4x'_2 \geq 1$$

$$6x'_1 + 3x'_2 \geq 1$$

$$3x'_1 + 6x'_2 \geq 1$$

$$x'_1 + 5x'_2 > 1$$

$$x'_1, x'_2 \geq 0$$

- B is a minimizing player.
- He wishes to minimize v , i.e.,

$$\text{Max } \frac{1}{v} = y'_1 + y'_2 + y'_3 + y'_4$$

subject to

$$5y'_1 + 6y'_2 + 3y'_3 + y'_4 \leq 1$$

$$4y'_1 + 3y'_2 + 6y'_3 + 5y'_4 \leq 1$$

$$y'_1, y'_2, y'_3, y'_4 \geq 0$$

- As can be observed that both these problem are primal dual pair.
- If we solve any one of them, the solution to the other can be obtained.

- Solving the second one (Problem for B) using simplex method we get $y'_1 = 0, y'_2 = \frac{4}{27}, y'_3 = 0, y'_4 = \frac{1}{9}$ and $\frac{1}{v} = \frac{7}{27}$ implies $v = \frac{27}{7}$.

Thus, **B's optimal strategy** is:

$$(y_1, y_2, y_3, y_4) = (0, \frac{4}{7}, 0, \frac{3}{7})$$

- Using the optimal table of the dual problem (B) we get the optimal primal solution (A) as

$$x'_1 = \frac{2}{27}, \quad x'_2 = \frac{5}{27}$$

with same optimal value for $\frac{1}{v}$ or v .

Thus, **A's optimal strategy** is:

$$(x_1, x_2) = (\frac{2}{7}, \frac{5}{7})$$

- As we added 3 to each pay-off in the original table, therefore the **optimal value of the game** is:

$$v - 3 = \frac{6}{7}.$$